[INCH-POUND]
GG-T-356B
30 June 1998
SUPERSEDING
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25 September 1967

FEDERAL SPECIFICATION

THERMOMETER, SELF-INDICATING, LIQUID-IN-GLASS, MINIMUM (METEOROLOGICAL)

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for two types and ten classes of organic liquid-in-glass, low temperature, complete immersion thermometers.

1.2 Classification. The thermometers shall be of the following types and classes. The type and class to be furnished shall be as specified (see 6.2).

Type I - Celsius (Centigrade) scale

Class 1: -20 °C to +60 °C
Class 2: -30 °C to +50 °C
Class 3: -40 °C to +45 °C
Class 4: -70 °C to +35 °C
Class 5: As specified in the contract or purchase order (not to exceed a span of 100 °C).

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 6660

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Type II - Fahrenheit scale

Class 6: -5 °F to +135 °F
Class 7: -20 °F to +120 °F
Class 8: -40 °F to +110 °F
Class 9: -90 °F to +90 °F
Class 10: As specified in the contract or purchase order (not to exceed a span of 180 °F).

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

MILITARY STANDARD

MIL-STD-2073-1 - DoD Material Procedures for Development and Application of Packaging Requirements

FEDERAL PUBLICATION

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

Monograph 150 - Liquid-in-Glass Thermometry

(Activities outside the Federal Government may obtain copies of federal specifications, standards and commercial item descriptions as specified in the General Information section of the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index is for sale on a subscription basis from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Single copies of this specification, and other federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC, 20407.)

(Federal Government activities may obtain copies of federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions form established distribution points in their agencies.)

(Copies of military standards required by contractors in connection with specific procurement functions are obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packing (DoD Adopted)

(Private sector and civil agencies may purchase copies of this voluntary standard from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

(DoD activities may obtain copies of those adopted voluntary standards listed in the DoD Index of Specifications and Standards free of charge from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Toxic chemicals, hazardous substances, and ozone depleting substances (ODS). The use of toxic chemicals, hazardous substances, or ODS shall be avoided, whenever feasible.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Performance characteristics. The thermometer shall be a solid stem, organic liquid-in-glass thermometer. Class 1 thermometers shall have a Celsius (Centigrade) scale designation, and class 2 thermometers shall have a Fahrenheit scale designation. The thermometer shall meet all the requirements of the NIST Monograph 150 for organic liquid-in-glass, complete immersion thermometers. The supplier shall supply a certificate of calibration as proof of calibration in accordance with NIST Monograph 150 for organic liquid-in-glass, complete immersion thermometers.

3.4.1 Scale. The scale shall be in 1 degree graduations.

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3.4.2 Expansion chamber. The expansion chamber shall be set at 57 °C (135 °F) or 71 °C (160 °F), as specified in the contract or purchase order.

3.4.3 Stem markings. Each solid stem shall be marked near the upper end and to the right of the scale with its own unique serial number. A symbol of the procuring agency shall precede the serial number. (Example: Defense Supply Center Richmond will be DSCR-Serial No.----.) The serial number(s) will be designated when the purchase order or contract is placed.

3.4.4 Backing. The thermometer shall be permanently fixed to a backing. The backing shall be no more than 2.30 cm (29/32 in) wide and 30.5 ± 0.2 cm (12 ± 1/16 in) long. There shall be a 0.3 cm (1/8 in) hole which shall be suitable for use as a mounting hole. The 0.3 cm (1/8 in) hole shall be centered and positioned within 0.6 cm (1/4 in) of the end of the backing adjacent to the expansion chamber. The backing shall be made of corrosion resisting material.

3.4.4.1 Mounting. The thermometer shall be laterally centered on the backing and permanently mounted to it. The method of mounting shall not interfere with the readability of the scale and serial number. The bulb shall not extend beyond one end of the backing, and the expansion chamber shall not block the backing's mounting hole.

3.4.5 Index. The index, which shall be used to provide for self-registration, shall be made of black glass and shall be loosely fitted into the bore of the stem. The index shall be slightly longer than the diameter of the bulb. The index shall be straight, smoothly formed, and provided with rounded terminal knobs. The index shall retreat with the organic liquid when the temperature falls, and shall permit the organic liquid to flow past it when the temperature rises.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. First article inspection shall consist of all the examinations and tests specified herein on five preproduction samples when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.

4.4 Quality conformance inspection. Each thermometer assembly shall be examined for conformance with the requirements specified in 3.4. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations, dimensional measurements and functional tests. Noncompliance with any specified requirements or presence of one or more defects shall constitute cause for rejection.

4.4.1 Visual and dimensional inspection. A visual inspection shall be made of the backings. Any thermometer whose backing has imperfect stamping shall be rejected. The backs shall be measured for conformance with 3.4.4. Backings which do not conform shall be rejected. Backings which are formed or bent in any manner, which shall transfer any undue strain to the thermometer tubes when the components are properly assembled, shall be rejected. A visual inspection shall be made of the thermometer's glass stem, bulb, expansion chamber, index and scale for conformance with 3.4. Thermometers with glass stems, bulbs, expansion chambers, indices and scales which do not conform shall be rejected.

4.4.2 Functional. A functional test will be conducted in accordance with NIST Monograph 150. The thermometer shall be accepted or rejected in accordance with criteria set forth in NIST Monograph 150.

4.5 Test methods.

4.5.1 Dimensional. All dimensional inspection shall be made with a ruler capable of being read to the nearest millimeter (1/16 inch).

4.5.2 Functional. The methods of inspection will be in accordance with NIST Monograph 150.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements shall be in accordance with ASTM D 3951 or when applicable the levels of protection conform to level A, level B or level C as defined in MIL-STD-2073-1, unless detailed requirements are specified in the contract.

6. NOTES

INFORMATION FOR GUIDANCE ONLY. This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.

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6.1 Intended use. The organic liquid-in-glass, low temperature, complete immersion thermometers covered by this specification are intended for use in the measurement of minimum temperatures for meteorological purposes.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class and Type or temperature range of thermometer (see 1.2).
c. Set point for the expansion chamber: 57 °C (135 °F) or 71 °C (160 °F) (see 3.4.2).
d. Stem markings. The symbol of the procuring agency and the starting serial number. (Example Defense General Supply Center Richmond will be DSCR-Serial No.----). (See 3.4.3).
e. Unit quantity required.
f. Whether a first article is required (see 3.1).
g. Packaging requirements (see 5).

6.3 Definitions.

a. Bulb: The reservoir for the thermometer liquid. The bulb of a thermometer will contain a volume equivalent to a specific number of degrees of the scale depending on the coefficient of expansion of the thermometric liquid and bulb glass.
b. Stem: The glass capillary tube through which the thermometric liquid advances or retreats with changes in temperature.
c. Scale: The scale graduated in degrees or multiples or submultiples of degrees.
d. Expansion chamber: An enlargement at the top end of the capillary bore having a volume equivalent to not less than the volume of a 20 mm length of unchanged capillary. Smaller chambers are not regarded as expansion chambers.
e. Complete immersion thermometer: A thermometer designed to indicate temperatures correctly when the whole thermometer, including expansion chamber, is subjected to the temperature being measured.

6.4 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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Type: 1 - Type I, Celsius (Centigrade) Scale
2 - Type II, Fahrenheit Scale

Class: 1 - -20 °C to +60 °C
2 - -30 °C to +50 °C
3 - -40 °C to +45 °C
4 - -70 °C to +35 °C
5 - As specified in the contract or purchase order (not to exceed a span of 100 °C).
6 - -5 °F to +135 °F
7 - -20 °F to +120 °F
8 - -40 °F to +110 °F
9 - -90 °F to +90 °F
10 - As specified in the contract or purchase order (not to exceed a span of 180 °F).

Specification number

6.5 Subject term (key word) listing.

Low temperature
Total immersion

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:
Custodians
Army - CR1
Navy - OS
Air Force - 99

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FCGC

Preparing Activity:
DLA - GS

Reviewers
Army - CR4
Navy - AS
Air Force - 01

(Project 6660-0123)

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